Band-1 Jacobian Conjecture Dim-2: Special Maps Probe (Collapse 2.11)

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Band-1 Jacobian Dim-2 — Special Maps (Lit + Statement Lock)

Honesty stamp. This note does **not** claim the Jacobian conjecture solved in dimension 2. It does **not** claim the Jacobian conjecture solved in all dimensions. It does **not** claim a Clay-millennial solution or a Theory of Everything. **GATE-04** is locked out of scope. **2026 higher-n counterexample announcements are treated as CLAIMS only** (dash rule); they are **not** elevated here and **do not** close JC-2.

> **Locked OPEN:** full complex JC-2 (unrestricted Keller maps) — **OPEN** (Band 3-4).
> **Special-closed (literature):** bounded-degree / tame-triangular / degree-pattern / line-injectivity / rational-inverse classes (see ledger).
> **Real JC: FALSE** (Pinchuk 1994) — orthogonal to complex JC-2.

Abstract

Keller (1939) asked whether a polynomial map $F\colon K^n \to K^n$ with $\det JF$ a nonzero constant (a **Keller map**) must be a polynomial automorphism. The classical focus is $n = 2$ over $\mathbb{C}$. **Full JC-2 remains open.** Many **special structured** subclasses are settled: degree ≤ 2 (Wang); degree ≤ 100 (Moh 1983); degree ≤ 104 (Nguyen, *Quaestiones Mathematicae* 2025, improving Moh); Appelgate-Onishi degree-factor patterns; Gwozdziewicz line-injectivity; maps with rational inverse; and the elementary/triangular generators of the **tame** automorphism group of $\mathbb{A}^2$ (Jung-van der Kulk). Reductions of the general conjecture (Bass-Connell-Wright / Yagzhev cubic homogeneous; Druzkowski cubic-linear) organize the landscape but do not close JC-2.

Box: optional sympy Jacobian checks on a few structured dim-2 examples (Jac $= 1$ for tame/triangular toys; nonconstant Jac for a non-Keller control). **Not** a proof.

Problem (collapse Band-2 item 2.11 → Band-1 lit+statement)

1. Ledger Keller JC; dim-2 status; known **special** classes.
2. Lock: full JC-2 **OPEN** vs special subclasses **literature-closed**.
3. Optional tiny Jac sanity checks. Keep Clay / ToE / GATE-04 / "JC solved" off the claim surface.
4. Treat **2026 higher-n claims** as **claims** only.

Literature ledger (web check; not re-proved on box)

Year	Result	Authors	Status
1939	Keller problem / JC stated ($n = 2$ integer coeffs)	Keller	classical statement

Year	Result	Authors	Status
classical	$\mathrm{Aut}(\mathbb{C}^2)$ generated by affine + elementary (tame)	Jung-van der Kulk	**structure theorem** (not full JC-2)
1980	JC for $\deg F \leq 2$ (all n)	Wang	**proved**
1982	Reduction to cubic homogeneous	Bass-Connell-Wright; Yagzhev	**proved reduction**
1983	JC-2 for deg ≤ 100	Moh	**proved** (computer-assisted)
1985	JC-2 when a component degree has ≤ 2 prime factors	Appelgate-Onishi	**proved** (special)
1994	Real JC false (positive Jac, not injective)	Pinchuk	**proved over** $\mathbb{R}$
~	Keller + injective on a line $\Rightarrow$ automorphism ($n = 2$)	Gwozdziewicz	**proved** (special)
2000	Survey monograph	van den Essen	secondary
2020	Frameworks for hypothetical 2D Keller maps	Borisov, *EJC*	combinatorial attack surface
2025	JC-2 through degree **104** (Newton polygon); extra special classes	**Nguyen**, *Quaestiones Mathematicae* 48	**literature proved** (bounded degree)
—	Full complex JC-2 unrestricted	—	**OPEN** (Band 3-4)
2026	$n \geq 3$ Keller counterexample announcements	Alpöge et al. (secondary)	**CLAIM only** (house rule); **does not close JC-2**

Canonical statement. A polynomial map $F = (F_1, \ldots, F_n)\colon \mathbb{C}^n \to \mathbb{C}^n$ with $\det JF \in \mathbb{C}^\times$ is called a Keller map; JC asserts every Keller map is a polynomial automorphism.

Canonical dim-2 degree citation. T.T. Moh, *On the Jacobian conjecture and the configurations of roots*, J. Reine Angew. Math. **340** (1983). Improvement: T. Nguyen, arXiv:1902.05923; *Quaestiones Mathematicae* **48** (2025), DOI 10.2989/16073606.2025.2482655.

Real caveat. Pinchuk (1994) kills the **real** Jacobian conjecture; complex JC-2 is a different theorem.

What is proved vs open (statement lock)

Special-closed / classical (cite carefully)

1. **Degree** ≤ 2 — Wang (all dimensions).
2. **Degree** ≤ 100 **(plane)** — Moh (1983).
3. **Degree** ≤ 104 **(plane)** — Nguyen (journal 2025); still **bounded-degree special**, not full JC-2.
4. **Degree-pattern specials** — Appelgate-Onishi (component degree a product of at most two primes).
5. **Line-injectivity special** — Gwozdziewicz: injectivity of F on some line $\Rightarrow$ automorphism.
6. **Rational-inverse class** — if a rational inverse exists, F is a polynomial automorphism.
7. **Tame / elementary / triangular maps** — generators of $\mathrm{Aut}(\mathbb{C}^2)$ (Jung-van der Kulk); such maps are automorphisms by construction. JC-2 is equivalent to asking whether every plane Keller map is tame.
8. **Reductions** — Bass-Connell-Wright / Yagzhev; Druzkowski (organize general JC; do not settle JC-2).
9. **Real JC** — **false** (Pinchuk).

Locked OPEN

1. **Full complex Jacobian conjecture in dimension 2** (unrestricted Keller maps) — **OPEN**. House band: **Band 3-4** (H-3.03 / GEO-JS).
2. **Keller ⇒ tame** in the plane — **OPEN** (equivalent form).

2026 higher-n (CLAIMS only)

Public 2026 announcements of explicit Keller maps in dimension $n = 3$ (and stabilizations $n \geq 3$) with constant nonzero Jacobian failing injectivity are **ledgered as CLAIMS / heat**. This house **does not** stamp them as closed theorems in this Band-1 probe, and **even if accepted elsewhere they do not close JC-2**. Dash rule: treat 2026 higher-N claims as claims.

Optional box illustration (not a proof)

Sympy Jacobian determinants on structured toys (`check_jacobian_examples.py`):

Example	Class	$\det JF$	Remark
(x, y)	identity	1	automorphism
$(x,\ y + x^2)$	elementary triangular	1	tame; inverse $(x,\ y - x^2)$
$(2x + 3y,\ 5x + 8y)$	affine SL_2	1	linear automorphism
$(x + (y + x^2)^2,\ y + x^2)$	composed elementaries	1	tame
$(x + y^3,\ y)$	triangular cubic	1	automorphism
$(x^2,\ y)$	negative control	$2x$	**not** Keller

Label: Jac sanity only. **Not** a search for counterexamples. **Not** a JC-2 proof.

Why Band-1 vs conjecture

- **Band-1 (this probe):** honest lit+statement lock for **special** dim-2 maps; optional Jac toys.
- **Band 3-4:** unrestricted JC-2.
- **Not claimed:** JC solved; JC-2 closed; Clay; ToE; elevation of 2026 higher-n claims.
- **GATE-04:** LOCKED.

Reproducibility

- Lab: `/workspace/mission-control/experiments/band1/jacobian_dim2_special/`
- Artifacts: `literature_ledger.json`, `results.json`, `SUMMARY.txt`, `METHOD.md`, `README.md`
- Box: `/workspace/.venv_jc/bin/python check_jacobian_examples.py`

Conclusion

Lit lock: Keller JC classical; **full complex JC-2 OPEN**; several **special structured** classes **literature-closed** (Wang / Moh / Nguyen deg<=104 / Appelgate-Onishi / Gwozdziewicz / rational-inverse / tame generators); Pinchuk kills **real** JC; 2026 higher-n items are **CLAIMS only**.
Box: Jac = 1 on tame toys; non-Keller control has nonconstant Jac.
Do not claim JC solved. No Clay / ToE. GATE-04 locked.